

Consolidated Azure Services Requirements

This document consolidates the **Azure service requirements for the Huella Digital platform**, based on the approved application assumptions and environment-specific assessments.

Each service required by the solution is listed below with its **key configuration parameters**, including **service tier, SKU, capacity sizing, usage volumes, and relevant options**.

The purpose of this document is to provide a **clear, unambiguous reference** for validating cloud readiness, cost estimation, and automated infrastructure provisioning via Infrastructure as Code.

Azure Static Web Apps (Frontend)

Service: Azure Static Web Apps

Plan: Standard

Custom Domain: Enabled

Authentication: Entra ID

Traffic:

- ~1,500–6,000 page views/month
- ~1 MB per page view (average)

Estimated Monthly Bandwidth: ~2–6 GB per month

Billing Model: Fixed monthly

Pricing Calculator Inputs:

- Tier: Standard
 - Apps: 1
 - Bandwidth: 6 GB
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Azure App Service (Backend API)

Service: Azure App Service (Linux)

Plan Tier: Standard S1

Region: Same as primary data region

Instance Count: 1

Autoscale: Disabled

OS: Linux

Runtime: .NET / Node / Python (as applicable)

Capacity:

- vCPU: ~1
- Memory: ~1.75–3.5 GB

Billing Model: Fixed monthly (App Service Plan)

Pricing Calculator Inputs:

- Operating System: Linux
 - Tier → Standard S1
 - Instance Type: S1 (1 Core, 1.75 GB RAM, 50GB Storage)
 - Instances: 1 × 730 Hours
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Azure Functions

Service: Azure Functions

Plan: Consumption

Runtime: Python

Region: Same as backend

Execution Model: Event-driven

Usage Estimates (Monthly):

- Total executions: ~6,400

- Average execution duration: 5–40 seconds
- Memory:
 - Background tasks: 512 MB
 - AI orchestration: 1–2 GB

Billing Model: Pay-per-execution

Pricing Calculator Inputs:

- Tier: Consumption
 - Memory Size: 1024 MB
 - Execution time: 20.000 ms
 - Executions: ~6,400/month
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Azure Database for PostgreSQL

Service: Azure Database for PostgreSQL – Flexible Server

Tier: Burstable

SKU: B2ms

vCores: 2

Memory: 8 GB

Storage:

- Type: Premium SSD
- Size: 32 GB
- Provisioned IOPS: 0 (default)

High Availability: Disabled

Backup:

- PITR enabled
- Redundancy: LRS
- Backup storage: 32 GB

Billing Model: Compute + storage (hourly)

Pricing Calculator Inputs:

- Deployment option: Flexible Server
 - Tier: Burstable
 - Compute
 - Type: B2MS (2 vCore)
 - Servers: 1 × 730 hours
 - Storage
 - Tier: Premium SSD
 - Size: 32 GB
 - **Backup Storage**
 - Redundancy: LRS
 - Size: 32 GB
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Azure Storage Account

Service: Azure Blob Storage

Account Type: General Purpose v2 (GPv2)

Hierarchical Namespace: Enabled (ADLS Gen2)

Redundancy: LRS

Access Tier: Hot

Capacity Target: 50 GB

Soft Limit Alert: 40 GB

Estimated Monthly Operations:

- Writes: ~10,000
- Reads: ~20,000
- Iterative reads/writes: ~3,000

- Data write: ~30 GB
- Data retrieval: ~10 GB

Billing Model: Pay-as-you-go

Pricing Calculator Inputs:

- Type: Block Blob Storage
 - Performance: Standard
 - Storage Account Type: General Purpose V2
 - File structure: Hierarchical Namespace
 - Access tier: Hot
 - Redundancy: LRS
 - Capacity
 - Size: 50 GB
 - Saving Options
 - Pay as you go
 - Operations and Data Transfer
 - Write operations: 10k
 - Read operations: 20k
 - Iterative read operations: 3k
 - Iterative write operations: 3k
 - Data retrieval: 10 GB
 - SSH File Transfer Protocol: Disabled
 - All other operations: 0
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Azure Key Vault

Service: Azure Key Vault

Tier: Standard

Managed HSM: No

Stored Items:

- Secrets: ~8–15
- Keys: 0–1 (software)
- Certificates: 0–1

Operations:

- ~200–300/month

Billing Model: Per-operation

Pricing Calculator Inputs:

- Vaults
 - Standard Operations: ~300/month
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Azure Monitor (Log Analytics + App Insights)

Services:

- Application Insights
- Log Analytics Workspace

Log Retention: 30 days

Sampling: 10–25%

Estimated Ingestion (Optimized):

- Total: ~0.5–1.5 GB/month
 - Analytics Logs: ~0.5–1 GB
 - Basic Logs: ~0.3–0.5 GB

Billing Model: Ingestion-based

Pricing Calculator Inputs:

- Log Analytics ingestion:
 - Basic Logs: ~ 0.5 GB/month

- Analytics Logs: ~ 1 GB/month
 - **Interactive Retention:** 30 days
 - Query:
 - Daily queries: 50
 - Data scanned per query: 1 GB
 - Application Insights:
 - Data ingestion: 0.05 GB/day
 - Data retention: 1 month
 - Standard web test: 1 every 5 min for 730 hours/month
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Azure Communication Services (Email)

Service: Azure Communication Services – Email

Billing Model: Pay-as-you-go

Estimated Monthly Volume:

- Emails: ~300–800
- Avg payload size: up to 5 MB

Features Used:

- Transactional email only
- No SMS / Voice / Chat

Pricing Calculator Inputs:

- Emails
 - Sent: 500/month
 - Size: 5 MB/mail
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Azure AI Search

Service: Azure AI Search

Tier: Basic

Search Units: 1

Partitions: 1

Replicas: 1

Indexed Data:

- Documents: ~100–500
- Estimated chunks: ~50k–300k

Billing Model: Fixed monthly

Pricing Calculator Inputs:

- Tier: Basic
 - Search Units: 1 × 730 hours
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Azure OpenAI Service

Service: Azure OpenAI

Model: GPT-4o-mini

Deployment Type: Standard (On-demand)

Throughput Reservation: None

Usage Estimates (Monthly):

- Requests: ~1,000
- Tokens per request: ~9,500
- Total tokens: ~9.5M/month

Billing Model: Token-based consumption

Pricing Calculator Inputs:

- Model type: Language Models
- Model: GPT-4o-mini
- Pricing Strategy: standard (on-demand)

- Deployment type: Global
 - Input tokens: 8 M/month
 - Cached input tokens: 8M/month
 - Output tokens: 1.5 M/month
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Final Notes for Cost Modeling

- All services are **right-sized for STAGING**
- No autoscaling-heavy or premium SKUs
- Flat-cost services dominate (predictable billing)
- Usage-based services are capped and observable
- Architecture preserves **full parity with PROD**